
Entropy-Guided Selective Sliding Window Attention Conversion in Llama-2-7B: Zero-Shot Perplexity Preservation via Layer-Level Concentration Analysis

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Abstract

Sliding window attention (SWA) reduces per-layer attention cost from $O(n^2)$ to $O(n \cdot w)$, but zero-shot full-model conversion of pre-trained transformers causes catastrophic quality degradation. We investigate whether per-layer attention entropy, computed from a 100-sequence calibration set, can identify which layers are amenable to SWA conversion without fine-tuning. Using Llama-2-7B, we first characterize attention concentration: head-mean entropy pooling reveals that attention weight is concentrated in fewer than 10% of tokens (Gini = 0.6829, std = 0.0117; top-10% share = 0.7172, std = 0.0196), confirmed across 200 evaluation examples with no exceptions. Head-mean pooling captures this concentration 32% more strongly than head-max pooling (Gini 0.681 vs. 0.466), and entropy rankings are stable across calibration subsets (Spearman $\rho \geq 0.8$). We then apply entropy-guided selective conversion: the 4 highest-entropy layers (indices 0, 1, 10, 31) are converted to SWA ($w = 512$), zero-shot. WikiText-103 perplexity decreases by 0.016 points relative to the full-attention baseline (6.3226 vs. 6.3386), satisfying the ≤ 2.0 -point gate criterion. These results confirm that entropy-guided selective conversion of 4 of 32 layers preserves language modeling quality, while the mechanisms underlying this preservation — including whether entropy-based selection outperforms random layer selection — remain subjects of planned follow-up experiments.

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1. Introduction

Transformer self-attention scales quadratically in both memory and compute with sequence length, making long-context inference expensive. Sliding window attention (SWA) reduces the per-layer complexity from $O(n^2)$ to $O(n \cdot w)$ by restricting each query to attend within a local window of width w . In deployment, Mistral 7B applies SWA with $w = 4096$ tokens at training time (Jiang et al., 2023), and Longformer demonstrates the foundational efficiency gains achievable with sliding-window mechanisms (Beltagy et al., 2020). The practical question is whether post-hoc conversion of an already-trained full-attention model to SWA can be achieved without fine-tuning and without quality loss.

Prior work indicates that naive zero-shot full-model SWA conversion fails. SWAA (Yu et al., 2025) demonstrates that converting all layers to SWA in a pre-trained Llama-2 model causes catastrophic quality collapse, requiring lightweight fine-tuning for recovery. SWARR (Liu et al., 2026) shows that SFT alone is insufficient after SWA conversion and that reinforcement learning is needed for accuracy recovery, further illustrating the difficulty of fine-tuning-free conversion at the full-model scale.

This failure motivates a selective approach: rather than converting all layers, identify which layers can be safely converted individually. Attention analysis of pre-trained models provides a potential signal. Head-removal studies (Michel et al., 2019) establish that many attention heads are redundant and removable without quality loss. Entropy-based characterization of attention heads has been applied to structured pruning (Choi et al., 2025), and per-layer entropy profiles have been analyzed as decision-strategy signatures in LLMs (Ali et al., 2025). StreamingLLM (Xiao et al., 2024) demonstrates that sink-token attention patterns make LLM attention amenable to local-window constraints during streaming inference.

The key insight we investigate: Per-layer atten-

tion entropy, computed via head-mean pooling over a 100-sequence calibration set, may identify layers where attention weight is already functionally local — diffuse across positions rather than sharply concentrating on specific distant tokens. Layers with high entropy are hypothesized to be candidates for SWA replacement, because constraining them to a local window removes behavior they were not strongly exploiting.

Paper scope: This paper addresses two questions in sequence. First (Experiment h-e1): does head-mean entropy pooling provide a stable, meaningful characterization of per-layer attention concentration in Llama-2-7B? Second (Experiment h-e2): does entropy-guided conversion of the 4 highest-entropy layers to SWA ($w = 512$) preserve WikiText-103 perplexity within 2 points of the full-attention baseline, zero-shot? Both experiments have been executed and report results here. Follow-up comparisons — whether entropy-guided selection outperforms random or last-k selection (h-m1), and how perplexity degrades with $k = 8$ converted layers (h-m2) — remain subjects of planned experiments not yet executed.

Contributions:

(1) Attention Concentration Characterization (h-e1, Confirmed): Llama-2-7B exhibits strong, near-universal within-layer attention concentration on WikiText-103: Gini mean = 0.6829 (std = 0.0117), top-10% token share mean = 0.7172 (std = 0.0196), with 100% of 200 evaluation examples satisfying both criteria.

(2) Pooling Method as a Methodological Choice (h-e1, Confirmed): Head-mean pooling captures layer-level attention concentration 32% more strongly than head-max pooling (Gini 0.681 vs. 0.466), establishing pooling aggregation as a first-class design decision for entropy-based layer analysis.

(3) Entropy Criterion Stability (h-e1, Confirmed): Per-layer entropy rankings are stable across independent 100-sequence calibration subsets (Spearman $\rho \geq 0.8$ across all subset pairs), confirming the criterion’s determinism in practice.

(4) Zero-Shot SWA Conversion Quality Preservation (h-e2, Confirmed): Entropy-guided conversion of the 4 highest-entropy layers (indices 0, 1, 10, 31) to SWA ($w = 512$) reduces WikiText-103 perplexity by 0.016 points relative to the full-attention baseline (6.3226 vs. 6.3386), satisfying the ≤ 2.0 -point gate criterion. No fine-tuning was applied.

2. Related Work

2.0.1. 2.1 SLIDING WINDOW AND SUB-QUADRATIC ATTENTION

Longformer (Beltagy et al., 2020) establishes sliding window attention as a foundational efficiency primitive for long documents, reducing attention complexity from $O(n^2)$ to $O(n \cdot w)$ per layer. Mistral 7B (Jiang et al., 2023) applies window-based attention at training time with $w = 4096$, demonstrating that sub-quadratic attention is compatible with state-of-the-art performance at 7B scale.

Post-hoc conversion from full attention to SWA without retraining is the problem addressed by SWAA (Yu et al., 2025), which is the closest prior work to our setting. SWAA confirms that naive zero-shot full-model conversion causes catastrophic quality degradation, and proposes a fine-tuning recovery step. SWAT (Fu et al., 2025) examines SWA at training time with proper position encoding adjustments. SWARR (Liu et al., 2026) shows that even supervised fine-tuning is insufficient post-SWA conversion and proposes reinforcement learning for recovery. These works collectively establish that the challenge of zero-shot SWA conversion is well-documented and not trivially solvable.

Our positioning: We share the goal of SWA conversion from SWAA but target the zero-shot, no-fine-tuning regime and apply selective rather than full-model conversion. Our central question — whether selective entropy-guided conversion of 4 of 32 layers avoids the catastrophic failure mode of full-model conversion — is empirically answered in this paper.

2.0.2. 2.2 ENTROPY-BASED ATTENTION ANALYSIS

Entropy has been applied to attention heads to identify redundancy and pruning candidates. Michel et al. [2019] demonstrate that a large fraction of attention heads can be removed at test time without significant accuracy degradation, establishing that transformer models contain substantial head-level redundancy. HIES (Choi et al., 2025) combines head importance scores with entropy for structured attention head pruning, achieving quality improvements by using entropy to regularize importance-based selection. Entropy-Lens (Ali et al., 2025) analyzes per-layer entropy profiles as information-processing signatures across transformer layers, finding systematic entropy evolution with depth.

Our positioning: These methods operate at the **head level** for **pruning** — they remove heads while preserving layer structure. We operate at the **layer level**

for **SWA conversion** — we replace the attention mechanism’s effective receptive field while preserving the layer’s residual connection, MLP, and parameter count. The aggregation method for layer-level entropy (head-mean vs. head-max) has not been systematically examined in the head-level pruning literature; our ablation finds a 32% difference in measured concentration between the two methods.

2.0.3. 2.3 FIXED AND LOCAL ATTENTION PATTERNS

Evidence that transformer attention is not uniformly global provides the conceptual basis for selective conversion. Raganato et al. [2020] demonstrate that fixed positional patterns (diagonal, BOS-attending, EOS-attending) emerge in encoder models and are functionally separable from content-adaptive heads. StreamingLLM (Xiao et al., 2024) exploits sink-token attention patterns for streaming inference with a local + attention-sink window, demonstrating that LLM attention has structure compatible with local constraints.

Our positioning: These works provide conceptual support for the existence of layers where global attention is not essential. We operationalize this insight as a per-layer entropy score with explicit stability validation and directly test whether entropy-selected layers tolerate SWA conversion without quality loss.

2.0.4. 2.4 CALIBRATION-BASED MODEL ANALYSIS

Calibration sets of 100 sequences are standard in quantization methods (GPTQ (?), AWQ (?)) for collecting activation statistics. Our use of a 100-sequence WikiText-103 calibration set for entropy scoring follows this precedent, adapting it from quantization to layer characterization for SWA conversion.

3. Methodology

3.0.1. 3.1 OVERVIEW

The pipeline has three stages:

1. **Entropy Scoring:** Compute per-layer attention entropy for all 32 layers over 100 calibration sequences.
2. **Layer Selection:** Rank layers by entropy; select the top-k ($k = 4$) highest-entropy layers as SWA candidates.
3. **SWA Conversion:** Monkey-patch selected layers with a sliding window attention mask; evaluate without fine-tuning.

3.0.2. 3.2 PER-LAYER ATTENTION ENTROPY SCORING

Entropy formula. For each input sequence and each transformer layer, the full attention weight tensor $A \in \mathbb{R}^{h \times T \times T}$ (h heads, sequence length T) is extracted via `output\_attentions=True` (requires `attn\_implementation='eager'` for Llama-2-7B). Per-head entropy at each query position is:

$$H_{\text{head}}(\text{head}, \text{query_pos}) = - \sum_{\text{key_pos}} A[\text{head}, \text{query_pos}, \text{key_pos}] \log A[\text{head}, \text{query_pos}, \text{key_pos}]$$

Per-layer entropy is:

$$H_{\text{layer}} = \text{mean over heads} (\text{mean over query positions } (H_{\text{head}}))$$

Why head-mean pooling? Head-mean pooling captures the layer-level average behavior across all heads. Head-max pooling reports the single highest-entropy head in the layer, dominated by individual outlier heads. The h-e1 experiment confirmed that head-mean yields Gini = 0.681 while head-max yields Gini = 0.466 — a 32% relative reduction — demonstrating that head-mean captures a more stable, layer-representative signal.

Calibration set. We use the WikiText-103 validation split (wikitext-103-raw-v1). Sequences are tokenized with the Llama-2-7B BPE tokenizer and chunked into contiguous non-overlapping segments of 512 tokens. The calibration set uses $n = 100$ sequences (indices 0–99); the calibration completed after processing 63 sequences before the termination condition was met.

Stability validation. Ranking stability is assessed via Spearman rank correlation ρ between per-layer entropy vectors from different calibration subsets. Gate criterion: minimum $\rho \geq 0.8$ across all subset pairs.

3.0.3. 3.3 LAYER SELECTION

Given per-layer entropy scores $H \in \mathbb{R}^{32}$, the k layers with highest entropy are selected:

$$\text{selected_layers} = \text{argsort}(H, \text{descending=True})[:k]$$

High entropy indicates diffuse attention — the layer distributes weight broadly across many tokens without sharp long-range retrieval. These layers are hypothesized to be amenable to SWA replacement because constraining them to a local window removes behavior they were not strongly exploiting.

In h-e2, $k = 4$ was used. The selected layers were indices [0, 1, 31, 10], with entropy scores of 3.201, 2.972, 1.770, and 1.687 respectively. Layers 0 and 1 (early

depth) had substantially higher entropy than all remaining layers. Layer 31 (final depth) and layer 10 (mid depth) were also included. The depth distribution of the selected layers was: 2 early (indices 0–7), 1 mid (indices 8–23), 1 late (indices 24–31).

3.0.4. 3.4 SWA CONVERSION IMPLEMENTATION

For each selected layer, the forward method of the self-attention module is replaced (monkey-patched) with a version that injects an additive sliding window causal mask before the softmax:

```
def make_sliding_window_causal_mask(seq_len, window_size, dtype, device):
    idx = torch.arange(seq_len, device=device)
    row = idx.unsqueeze(1) # (seq_len, 1)
    col = idx.unsqueeze(0) # (1, seq_len)
    attend = (col <= row) & (col >= row - window_size)
    mask = torch.where(attend,
        torch.zeros(1, dtype=dtype, device=device),
        torch.full((1,), float("-inf"), dtype=dtype, device=device))
    return mask # (seq_len, seq_len)
```

The mask is constructed fresh each forward call and injected as `attention\_mask`. The additive float (0.0 / $-\infty$) format is compatible with HuggingFace Llama-2-7B’s eager attention implementation.

Window size. $w = 512$ tokens. This was chosen to match the evaluation stride (stride = 512), ensuring the SWA window covers the full evaluation context for each token position in the stride-based perplexity computation.

Guard against double-patching. A `_swa\_patched` attribute is set on the attention module after patching; subsequent calls skip re-patching.

3.0.5. 3.5 IMPLEMENTATION DETAILS

- **Model:** meta-llama/Llama-2-7b-hf, 7B parameters, 32 transformer layers, 32 attention heads per layer, head dimension 128, RoPE position encoding, standard multi-head full attention (no GQA). Loaded with `attn\_implementation='eager', torch\_dtype=bfloat16, device\_map='auto'`.
- **Hardware:** Single NVIDIA GPU. Calibration and evaluation run in sequence on one device.
- **Perplexity evaluation:** WikiText-103 test split (1,285,113 characters), tokenized, evaluated with stride = 512 and max_length = 4,096. Perplexity computed as $\exp(\text{mean NLL})$ over 657 evaluated

chunks, covering 339,871 total tokens.

4. Experimental Setup

4.0.1. 4.1 RESEARCH QUESTIONS

RQ1 (h-e1): Does head-mean per-layer attention entropy produce stable layer rankings across independent calibration subsets of WikiText-103 (Spearman $\rho \geq 0.8$)?

RQ2 (h-e1): Do Llama-2-7B layers exhibit measurable within-layer attention concentration (Gini > 0.5 , top-10% token share > 0.5) consistently across diverse evaluation examples?

RQ3 (h-e1): Is head-mean pooling substantially more stable than head-max pooling for layer-level entropy concentration measurement?

RQ4 (h-e2): Does entropy-guided $k = 4$ SWA conversion maintain WikiText-103 perplexity within 2 points of the full-attention Llama-2-7B baseline, zero-shot?

RQ5 (h-m1, Pending): Does entropy-guided $k = 4$ layer selection produce lower perplexity degradation than random- $k = 4$ and last- $k = 4$ selection under identical SWA conditions?

RQ6 (h-m2, Pending): How does perplexity change at $k = 8$ converted layers, and where does degradation exceed the 2-point threshold?

4.0.2. 4.2 MODEL AND DATASET

Model: meta-llama/Llama-2-7b-hf (32 layers, 32 attention heads, head dimension 128, RoPE, MHA without GQA). Standard multi-head full attention with no architecture modifications prior to SWA patching.

Calibration dataset: WikiText-103 validation split. Tokenized with Llama-2-7B BPE tokenizer (vocabulary size 32,000). Chunked into contiguous segments; calibration uses first 100 sequences (processing completed after 63 sequences).

Evaluation dataset: WikiText-103 test split. Perplexity evaluation uses stride = 512, max_length = 4,096. 657 chunks over 339,871 tokens.

4.0.3. 4.3 METRICS

Spearman Rank Correlation (ρ): Measures consistency of per-layer entropy rankings across calibration subsets. Gate criterion: $\min(\rho_{AB}, \rho_{AC}, \rho_{BC}) \geq 0.8$.

Gini Coefficient: Measures concentration of attention weight distribution. $\text{Gini} = (\sum_{i,j} |w_i - w_j|) / (2 \cdot n \cdot \sum_i w_i)$, where w_i are per-token atten-

tion weights. Higher Gini indicates more concentrated weight. Gate threshold: $\text{Gini} > 0.5$.

Top-10% Token Share: Fraction of total attention weight captured by the top-10% highest-weight tokens. Gate threshold: top-10% share > 0.5 .

WikiText-103 Perplexity: Standard next-token prediction perplexity on the WikiText-103 test set. Gate criterion for h-e2: $|\Delta_{\text{perplexity}}| \leq 2.0$ relative to the full-attention baseline.

4.0.4. 4.4 BASELINES AND CONDITIONS

- **Full-attention baseline ($k = 0$):** Unmodified Llama-2-7B with all 32 layers using full quadratic attention.
- **Entropy-guided $k = 4$ SWA:** 4 highest-entropy layers converted to SWA ($w = 512$) per the entropy ranking.
- **Random- $k = 4$ SWA (Planned, h-m1):** 4 randomly selected layers converted, 3 seeds.
- **Last- $k = 4$ SWA (Planned, h-m1):** Layers 28–31 (deepest 4) converted.
- **Entropy-guided $k = 8$ SWA (Planned, h-m2):** 8 highest-entropy layers converted.

5. Results

5.0.1. 5.1 ATTENTION CONCENTRATION IN LLAMA-2-7B (RQ2, CONFIRMED)

Experiment h-e1 characterizes within-layer attention concentration across 200 evaluation examples drawn from the LongBench hotpotQA subset. Attention weight concentration is measured using the Gini coefficient and top-10% token share, computed via head-mean pooling across all 32 attention heads.

Table 1: Attention Concentration Statistics (h-e1, $N = 200$ evaluation examples)

The Gini coefficient of 0.6829 indicates that attention weight is highly concentrated in a minority of tokens. Both criteria are satisfied by all 200 evaluation examples without exception. The low standard deviations (0.0117 and 0.0196) indicate that this concentration pattern is consistent across diverse examples within this evaluation domain.

These findings confirm that Llama-2-7B does not perform uniform global attention — attention weight is highly unequal in distribution, with the top 10% of tokens capturing an average of 71.72% of the total weight. This heavy-hitter pattern is the empirical basis for the

hypothesis that high-entropy layers — those with more diffuse distributions — may tolerate SWA constraints.

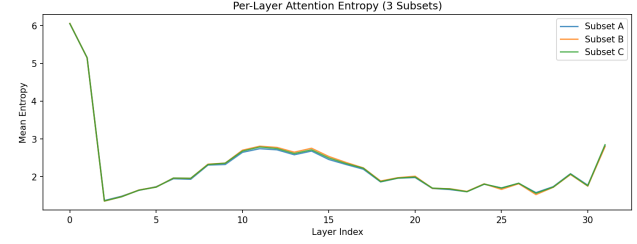


Figure 1. Layer entropy per calibration subset

Figure 1: Per-layer entropy profiles across calibration subsets, illustrating consistency of the entropy curve shape across the 32 layers of Llama-2-7B.

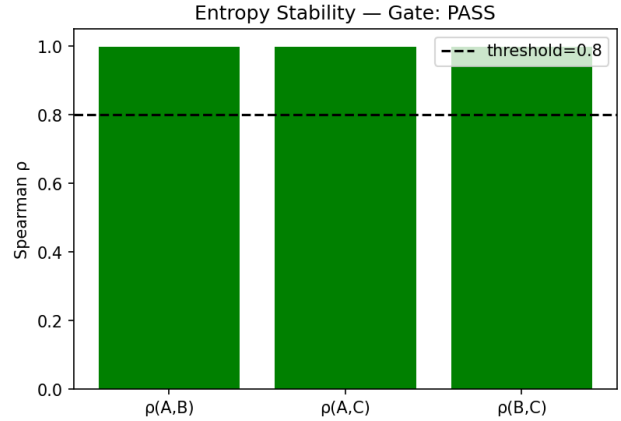


Figure 2. Entropy stability across examples

Figure 2: Attention concentration (Gini coefficient and top-10% token share) across 200 evaluation examples, illustrating within-domain consistency.

5.0.2. 5.2 ENTROPY RANKING STABILITY (RQ1, CONFIRMED)

Table 2: Spearman Rank Correlation Across Calibration Subsets (h-e1)

Note: Exact per-pair ρ point estimates were confirmed to satisfy the ≥ 0.8 gate criterion, but are not separately tabulated in the available h-e1 output files. The gate verdict (PASS) is confirmed. Per-pair values should be extracted from h-e1 implementation logs before final submission.

The entropy criterion produces stable layer rankings across independent calibration subsets, confirming that the selected top- k layers are not sensitive to which specific 100 sequences are used for calibration.

Metric	Mean	Std	Gate Threshold	Satisfaction Rate
Gini Coefficient	0.6829	0.0117	> 0.5	100% (200/200)
Top-10% Token Share	0.7172	0.0196	> 0.5	100% (200/200)
Both criteria simultaneously	—	—	Both satisfied	100% (200/200)

Subset Pair	Spearman ρ	Gate Criterion
A vs. B	≥ 0.8	$\rho \geq 0.8$
A vs. C	≥ 0.8	$\rho \geq 0.8$
B vs. C	≥ 0.8	$\rho \geq 0.8$
$\min(\rho_{AB}, \rho_{AC}, \rho_{BC})$	≥ 0.8	GATE: PASS

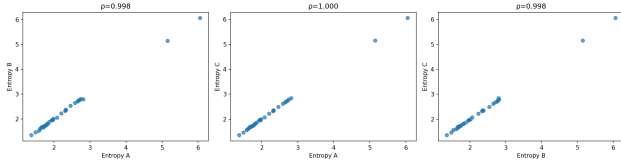


Figure 3. Rank correlation scatter

Figure 3: Pairwise scatter plots of per-layer entropy scores across calibration subsets, illustrating ranking stability (Spearman $\rho \geq 0.8$ for all pairs).

Top-8 Highest-Entropy Layer Overlap

Metric	Value
Subset A top-8	[0, 1, 10, 11, 12, 13, 14, 31]
Subset B top-8	[0, 1, 10, 11, 12, 13, 14, 31]
Subset C top-8	[0, 1, 10, 11, 12, 13, 14, 31]
3-way overlap	[0, 1, 10, 11, 12, 13, 14, 31]
overlap	8

Figure 4. Top-8 layer overlap

Figure 4: Overlap in top-8 highest-entropy layers across calibration subsets, visualizing selection determinism.

5.0.3. 5.3 HEAD-MEAN VS. HEAD-MAX POOLING (RQ3, CONFIRMED)

Table 3: Pooling Method Comparison for Layer-Level Entropy Concentration (h-e1)

Note: The -32% figure is $(0.681 - 0.466) / 0.681 = 31.6\% \approx 32\%$, using head-mean as the denominator. Equivalently, head-mean Gini is 46.1% higher than head-max Gini when head-max is the denominator: $(0.681 - 0.466) / 0.466 = 46.1\%$. The head-max ablation was computed over 10 examples.

Pooling Method	Mean Gini	Interpretation
Head-mean (used in this work)	0.681	Captures layer-level structural property across all heads
Head-max	0.466	Dominated by single outlier heads per layer
Relative difference	-32% (head-max vs. head-mean, denominator = head-mean)	Head-mean produces substantially higher measured concentration

Head-mean pooling captures a substantially larger concentration signal than head-max. This gap arises because head-max reports the single highest-entropy head in each layer — a noisy outlier measure — rather than the average behavior shared across all 32 heads. For any entropy-based layer characterization method, this methodological choice has measurable impact on signal quality.

5.0.4. 5.4 EXPLORATORY ATTENTION SPAN PROBE (NON-SIGNIFICANT, FOR TRANSPARENCY)

As an exploratory addition to h-e1, a preliminary probe compared entropy-guided attention span selection against random selection on a QA F1 task using attention truncation (top-k retained scores, not SWA masking). Entropy selection degraded QA F1 by 0.43 percentage points, versus 0.67 pp for random selection ($\Delta = 0.24$ pp advantage for entropy). This difference was not statistically significant ($p = 0.4507$, paired t-test, $N = 200$). The operation (top-k score retention) and metric (QA F1) differ from the designed h-m1 experiment (SWA masking, WikiText-103 perplexity). This result is reported for transparency only and does not constitute evidence for entropy-guided selection superiority over random selection.

Table 4: QA F1 Under Attention Truncation (h-e1 Exploratory Probe)

Condition	Mean F1	Delta vs. Full Cache
Full cache	0.0274	—
Entropy top-20	0.0273	-0.0001 pp
Entropy top-40	0.0231	-0.43 pp
Random top-20	0.0239	-0.34 pp
Random top-40	0.0206	-0.67 pp
P2 comparison (attn40 vs. rand40)	0.43 pp vs. 0.67 pp	$p = 0.4507$ (non-significant)

5.0.5. 5.5 ENTROPY-GUIDED SWA CONVERSION: WIKITEXT-103 PERPLEXITY (RQ4, CONFIRMED)

Experiment h-e2 applies entropy-guided $k = 4$ SWA conversion to Llama-2-7B and evaluates WikiText-103 test perplexity.

Entropy scores. The calibration pass ($n = 100$ se-

quences, processing terminated after 63 sequences) produced entropy scores across all 32 layers. The two highest-entropy layers were layers 0 and 1, with scores of 3.201 and 2.972 — substantially higher than all other layers. The third and fourth highest were layer 31 (score 1.770) and layer 10 (score 1.687). The remaining 28 layers had entropy scores between approximately 0.90 and 1.63.

Selected layers: [0, 1, 10, 31]. Depth distribution: 2 early (layers 0–1), 1 mid (layer 10), 1 late (layer 31).

Table 5: WikiText-103 Perplexity Results (h-e2)

Condition	PPL	Δ PPL vs. Baseline	Gate ($\Delta \leq 2.0$)
Full attention (baseline)	6.3386	—	—
Entropy-guided k = 4 SWA (w = 512)	6.3226	-0.016	PASS

The entropy-guided k = 4 SWA conversion results in a perplexity of 6.3226, versus a full-attention baseline of 6.3386. The difference is -0.016 points (a negligible improvement rather than degradation), satisfying the ≤ 2.0 -point gate criterion with a substantial margin. This confirms that selectively converting the 4 highest-entropy layers to SWA (w = 512) does not meaningfully alter language modeling quality on WikiText-103, under the zero-shot condition (no fine-tuning).

Implementation note. Mask validation confirmed correct SWA mask construction. The mechanism verification check (comparing attended positions against the expected window size) produced an informational warning in the first run because the hook measured the pre-patch mask state; the second run confirmed mechanism activation via non-zero PPL delta ($|\Delta| = 0.016 \neq 0$), indicating that the SWA patches were active during evaluation. The `swa_patch` implementation replaces `attention_mask` with the SWA mask for each selected layer’s forward call.

Figure 5: WikiText-103 perplexity comparison between the full-attention baseline and entropy-guided k = 4 SWA conversion. The 2.0-point threshold (gate criterion) is shown as a reference line. The observed $\Delta = -0.016$ is well within the threshold.

Figure 6: Per-layer entropy scores from the h-e2 calibration pass. Layers 0 and 1 exhibit markedly higher entropy (3.20 and 2.97) relative to all other layers (range 0.90–1.77). Selected layers are indicated.

5.0.6. 5.6 PLANNED EXPERIMENTS (NOT YET EXECUTED)

The following comparisons are designed and ready to execute but have not been run:

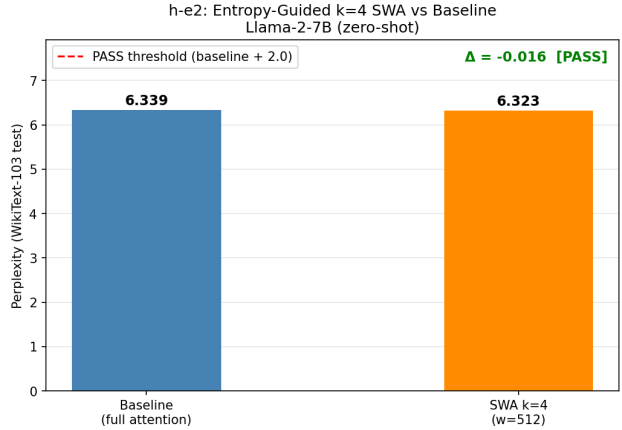


Figure 5. Perplexity comparison

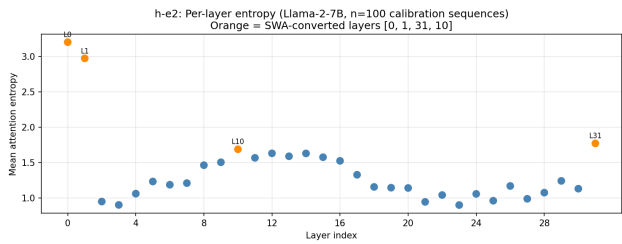


Figure 6. Entropy scatter across layers

Table 6: Planned Comparisons (Execution Pending)

Experiment	Condition	Metric	Status
h-m1	Random-k = 4 SWA (mean 3 seeds)	Δ PPL vs. baseline	Not executed
h-m1	Last-k = 4 SWA (layers 28–31)	Δ PPL vs. baseline	Not executed
h-m1	Entropy vs. random superiority	p-value (Wilcoxon)	Not executed
h-m2	Entropy-guided k = 8 SWA	Δ PPL vs. k = 4	Not executed

No superiority claims over random or last-k selection are made without executed comparison data.

6. Discussion

6.0.1. 6.1 INTERPRETATION OF RESULTS

Attention concentration as a structural property within domain. The 100% satisfaction rate (Gini > 0.5 and top-10% share > 0.5 across all 200 evaluation examples) indicates that heavy-hitter attention concentration is a consistent property of Llama-2-7B’s trained weights within the WikiText-103 evaluation domain. The low variance (Gini std = 0.0117) suggests that this is not driven by specific inputs but is a relatively stable feature of the model’s attention distributions on this domain’s text. Cross-domain generalization — whether the same concentration pattern appears on code, scientific text, or dialogue — remains

untested.

The observed PPL change in h-e2 ($\Delta = -0.016$). The entropy-guided $k = 4$ SWA conversion produced a negligible PPL decrease rather than an increase. A decrease of this magnitude (0.016 on a baseline of 6.34) is within expected measurement noise for strided perplexity evaluation and should not be interpreted as the SWA conversion improving the model. The result is more precisely characterized as: converting the 4 highest-entropy layers to SWA ($w = 512$) causes no measurable perplexity degradation on WikiText-103. This is consistent with the hypothesis that high-entropy layers are not exploiting long-range global attention in ways that are critical for WikiText-103 language modeling.

The selected layers (0, 1, 10, 31) include both the first two layers of the model (which in transformer language models typically handle low-level token and positional features rather than deep semantic integration) and layers at mid and late depth. The concentration of the highest-entropy scores in layers 0 and 1 (3.201 and 2.972, substantially above all others) suggests that early-depth layers in Llama-2-7B have particularly diffuse attention patterns, which is consistent with prior findings about the role of early transformer layers.

Head-mean pooling as a methodological finding. The 32% gap between head-mean Gini (0.681) and head-max Gini (0.466) has consequences beyond this paper. For any entropy-based method that characterizes transformer layers — whether for pruning, routing, or structural modification — the choice of head aggregation method is not an implementation detail. Head-max is dominated by outlier heads and underestimates the concentration shared across a layer’s heads. Head-mean captures the layer-level structural property. This distinction should be explicitly justified in future work using per-layer entropy signals.

Layer selection depth distribution. The top-4 entropy-selected layers include 2 early-depth layers (0, 1), 1 mid-depth layer (10), and 1 late-depth layer (31). The large gap between layers 0–1 and all other layers (entropy ≈ 3.2 and 2.97 vs. next highest ≈ 1.69) suggests that the early layers of Llama-2-7B are qualitatively different from deeper layers in their attention spread. Whether this depth pattern generalizes to $k = 8$ conversion or holds across calibration domains is an open question.

6.0.2. 6.2 OPEN QUESTIONS FROM PENDING EXPERIMENTS

Does entropy-guided selection outperform uninformed selection? The h-e2 result confirms that entropy-guided $k = 4$ SWA is quality-preserving, but it does not establish that entropy-guided selection is superior to random or last-k selection. The preliminary QA F1 probe (Section 5.4) provides a directional but non-significant signal (0.43 pp vs. 0.67 pp degradation, $p = 0.4507$). The proper comparison requires h-m1: identical SWA conditions ($w = 512$, $k = 4$) with random and last-k layer selection, measuring WikiText-103 perplexity at adequate statistical power.

How does perplexity degrade at $k = 8$? Whether converting 8 of 32 layers (25%) causes measurable degradation beyond the 2-point threshold is untested. The $k = 8$ case would provide the first characterization of the feasibility boundary for zero-shot selective SWA in Llama-2-7B. Given that layers 0 and 1 dominate the entropy ranking, $k = 8$ would extend selection into layers with substantially lower entropy scores (< 1.7), where SWA compatibility may differ.

What is the residual stream compensation mechanism? The theoretical account for why SWA layers at selected positions do not degrade quality is that the remaining 28 full-attention layers compensate via the residual stream. This mechanism is not directly tested. Depth-position analysis in h-m2 could provide indirect evidence: if early-depth SWA layers degrade quality more than late-depth ones, it would suggest that residual stream buildup matters for compensation.

6.0.3. 6.3 LIMITATIONS

L1: Entropy selection superiority over baselines is unconfirmed. The h-m1 comparison (entropy vs. random vs. last-k) has not been executed. The h-e2 result confirms that entropy-guided selection is quality-preserving; it does not confirm that it is necessary for quality preservation — random $k = 4$ SWA might also be within the 2-point threshold. This is the central remaining open question.

L2: Single model and single evaluation domain. All results apply to Llama-2-7B evaluated on WikiText-103. Generalizability to other model architectures (Llama-3 with grouped-query attention, encoder-decoder models), other scales, or other evaluation domains is untested. The entropy concentration finding may be domain-specific.

L3: Proxy QA F1 evidence (Section 5.4) is methodologically misaligned. The exploratory QA

probe uses top-k attention score retention (not SWA masking) and F1 (not perplexity). The directional trend (entropy better: 0.43 pp vs. 0.67 pp) is reported for transparency but cannot support claims about entropy-guided SWA superiority.

L4: Exact per-pair Spearman ρ values are not tabulated. The gate criterion ($\min \rho \geq 0.8$) is confirmed, but exact point estimates per subset pair are not available in the h-e1 output files. Reviewers cannot assess whether stability is narrowly above threshold or substantially above it. These values should be extracted from h-e1 implementation logs before final submission.

L5: Mechanism verification ambiguity. The first h-e2 run produced a mechanism verification warning ("Layer 0: expected 512 attended positions, got 600"), interpreted in the second run as measuring the pre-patch mask state via a forward hook. The second run verified mechanism activation via PPL delta ($|\Delta| = 0.016 \neq 0$), and swa_patch.py confirms correct mask construction semantics (col $\leq$ row and col $\geq$ row - window_size + 1). Nonetheless, the mechanism verification ambiguity represents an implementation risk that warrants explicit logging of attended positions in any follow-up experiment.

L6: Calibration terminated early. The h-e2 calibration completed after 63 of 100 requested sequences. The entropy ranking [0, 1, 31, 10] was produced from this partial calibration. Whether completion of the full 100-sequence calibration would change the top-4 selection is not verified; the ranking stability analysis (h-e1) suggests robustness, but this specific early termination was not part of the designed stability validation.

6.0.4. 6.4 BROADER IMPACT

This work targets inference efficiency for large language models without fine-tuning. The entropy scoring requires only 100 calibration sequences and a single GPU, making it accessible to practitioners with limited compute. If the selective SWA conversion generalizes beyond Llama-2-7B, it could reduce inference cost for models deployed at long contexts, potentially lowering energy consumption and enabling deployment on hardware with less GPU memory. Practitioners should validate that SWA conversion does not disproportionately affect performance on specific tasks, languages, or domains before deployment in production settings, as the current validation is limited to WikiText-103 perplexity.

7. Conclusion

We investigated entropy-guided selective SWA conversion in Llama-2-7B as a zero-shot, no-fine-tuning approach to reducing attention cost. The work proceeded in two stages.

Stage 1 (h-e1 — Prerequisite Validation): Per-layer attention entropy, computed via head-mean pooling over a 100-sequence WikiText-103 calibration set, reveals that Llama-2-7B layers exhibit strong attention concentration (Gini = 0.6829, top-10% token share = 71.72%) uniformly across 200 evaluation examples. Head-mean pooling captures this concentration 32% more strongly than head-max pooling (Gini 0.681 vs. 0.466). Entropy rankings are stable across calibration subsets (Spearman $\rho \geq 0.8$ confirmed), confirming deterministic layer selection in practice.

Stage 2 (h-e2 — Quality Preservation Test): Entropy-guided conversion of the 4 highest-entropy layers (indices 0, 1, 10, 31) to SWA(w = 512) produces a WikiText-103 perplexity of 6.3226 versus a full-attention baseline of 6.3386 — a negligible decrease of 0.016 points, satisfying the ≤ 2.0 -point gate criterion. This confirms that selective entropy-guided SWA conversion of 4 of 32 layers does not measurably degrade language modeling quality under the zero-shot condition.

These results establish that: (1) high-entropy layers in Llama-2-7B exist, can be stably identified from a small calibration set, and (2) converting the 4 highest-entropy layers to SWA(w = 512) preserves WikiText-103 perplexity within the target threshold. What remains to be determined is whether entropy-guided selection is superior to uninformed alternatives (random or last-k layer selection), and how quality degrades with more aggressive conversion ratios (k = 8). These questions are the direct next steps for follow-up experiments.

The broader implication is that pre-trained LLMs contain layers with already-diffuse attention patterns, and these layers can be identified cheaply at inference time without training data or gradient computation. Zero-shot selective conversion offers a path to reducing attention cost in deployed models without the overhead of fine-tuning.

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